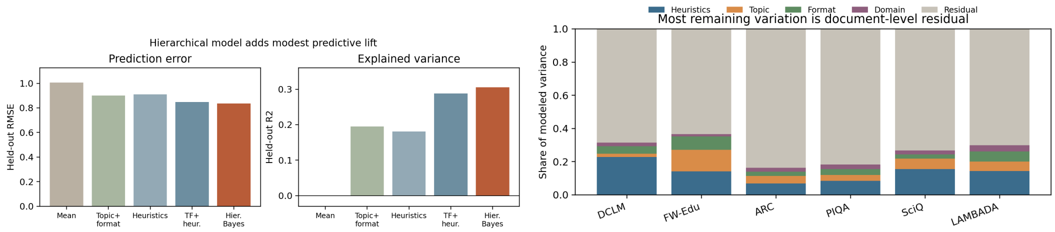

(a) Held-out predictive comparison. (b) Posterior variance decomposition.

Figure 1: The Bayesian hierarchy gives a modest but real predictive gain over fixed-effect baselines, and most unexplained variation remains document-level.

STATS 305C Applied Statistics III
Milestone 3: Model and Inference

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Goal and data. Milestone 2 showed that web-quality filters agree on some broad axes but still disagree substantially within topic. The main feedback was that a hierarchical model should do something actionable, not only restate this fact. We therefore fit a probabilistic model to six complete quality scores on a reproducible 120k-document sample from our 2.26M-document WebOrganizer subset, then use its posterior predictions to propose data-selection mixes. The modeled scores are DCLM, rounded FineWeb-Edu, and four perplexity-correlation filters (ARC-Easy, PIQA, SciQ, LAMBADA). Bounded scores are clipped at 10^{-4} and logit-transformed; FineWeb-Edu is treated as a continuous ordinal score. All responses and 17 heuristic covariates are standardized using the training split only. We use 96,097 training documents and 23,903 held-out documents, with 24 topics, 24 formats, and the top 500 registered domains plus an "other" bucket.

Generative model. For document i, let $y_i \in \mathbb{R}^6$ be the transformed standardized score vector and z_i be a design vector containing an intercept, standardized heuristics, topic indicators, format indicators, and domain indicators. We fit the multivariate Gaussian shrinkage model

$$y_i \mid B, \Sigma \sim \mathcal{N}_6(z_i^\top B, \Sigma), \tag{1}$$
$$B \mid \Sigma \sim \mathcal{MN}_{p,6}(0, V_0, \Sigma), \tag{2}$$
$$\Sigma \sim \mathcal{IW}(\nu_0, S_0). \tag{3}$$

This is a hierarchical random-effects model in which topic, format, and domain coefficients are partially pooled toward zero through block-specific prior scales. We use prior standard deviations 10 for the intercept, 1.5 for heuristics, 1 for topic and format effects, and 0.5 for domain effects; these regularize the many sparse domain levels while leaving common topics/formats flexible. The residual covariance Σ is intentionally full rather than diagonal because cross-filter disagreement is central to the scientific question. We set $\nu_0 = 10$ and $S_0 = I_6$.

Inference. The model is conjugate, so we implemented a blocked Gibbs sampler. With Z the training design matrix and Y the response matrix,

$$V_n = (Z^\top Z + V_0^{-1})^{-1}, \qquad M_n = V_n Z^\top Y,$$

and we alternate

$$B \mid \Sigma, Y \sim \mathcal{MN}(M_n, V_n, \Sigma), \quad \Sigma \mid B, Y \sim \mathcal{IW}(\nu_0 + n + p,\ S_0 + (Y - ZB)^\top (Y - ZB) + B^\top V_0^{-1} B).$$

We ran 4 chains, 300 iterations each, discarded 150 burn-in iterations, and kept every third draw, for 200 retained draws. Scalar diagnostics were stable: the worst $\widehat{R}$ was 1.025 and the minimum approximate ESS was 143.

Posterior analysis and baselines. Figure 1a compares against simpler baselines. Mean prediction has held-out RMSE 1.006 and $R^2 \approx 0$. Topic+format fixed effects reach $R^2 = 0.195$, heuristics reach $R^2 = 0.181$,

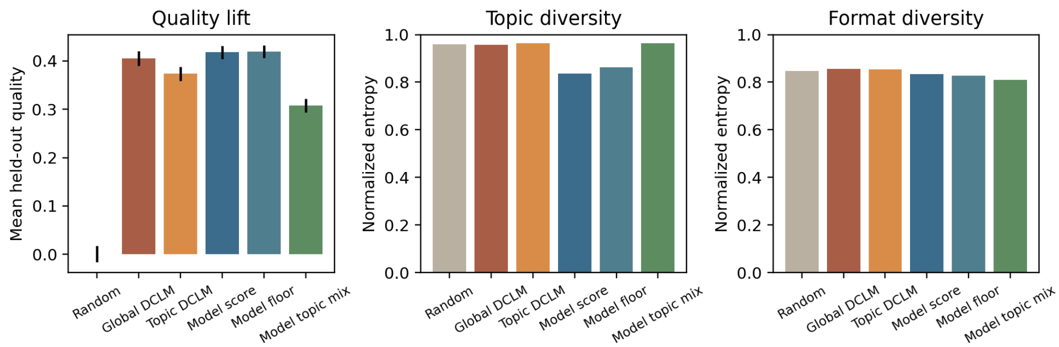

Figure 2: Actionable data-mix evaluation on held-out documents. All policies select the top 20% except the random reference. The model-based policy gives the highest composite quality but trades off some diversity.

and topic+format+heuristics reach $R^2 = 0.288$ with RMSE 0.847. The hierarchical model improves to RMSE 0.836 and $R^2 = 0.305$. This is not a dramatic gain, but it is consistent across scores and is largest for DCLM, FineWeb-Edu, SciQ, and LAMBADA. Figure 1b shows why the problem remains hard: after observed covariates, residual document-level variation explains roughly 63–84% of each score's modeled variance. Heuristics explain the largest non-residual share for DCLM (23%), SciQ (15%), and LAMBADA (14%); topic matters most for FineWeb-Edu (13%). The full residual covariance is useful: ARC-Easy and PIQA still have posterior residual correlation 0.86 after covariates, while DCLM and LAMBADA retain only 0.18 correlation. Thus benchmark-specific filters share signal that is not captured by topic, format, domain, or simple text heuristics.

Actionable data mix. We define composite held-out quality as the average of the six standardized scores and compare top-20% selection rules (Figure 2). Random selection has mean quality 0.001. A global DCLM threshold reaches 0.405 with normalized topic entropy 0.957. A within-topic DCLM threshold preserves topic entropy (0.964) but drops quality to 0.373. Ranking documents by posterior predicted quality gives the best quality, 0.418, but lower topic entropy (0.837). Our preferred actionable policy is a compromise: first keep a 5% within-topic floor by model score, then fill the remaining slots globally by model score. It attains quality 0.419, covers all 24 topics, and improves topic entropy to 0.862 relative to pure model ranking. The honest conclusion is that the model adds a small predictive and selection advantage over DCLM, but the gain comes with a diversity cost; a useful deployment should therefore expose the quality-diversity frontier rather than choose a single global threshold.

AI assistance. We used Codex and Claude Code for implementation assistance, report drafting, and debugging. The model, inference target, diagnostics, and interpretation were checked against locally generated artifacts and code committed in the repository.